MutantXL closure refutes the ECC2K-130 descent systems at degree 3–4, where XL needs 5–7

Results of `research/dreg_closure_mutantxl_20261016/`. The full text is `RESULTS.md`, and the run report is `reports/run-20261016T0235Z/`. Class **stage diagnostic**. The `n = 131` rows below are **derived extrapolations, not measurements**.

Question

Every refutation degree this repository recorded for the chained S_3 systems is XL's, and the ECC2K-130 cost model uses those degrees. Does MutantXL's fall closure at a fixed degree refute lower?

Result

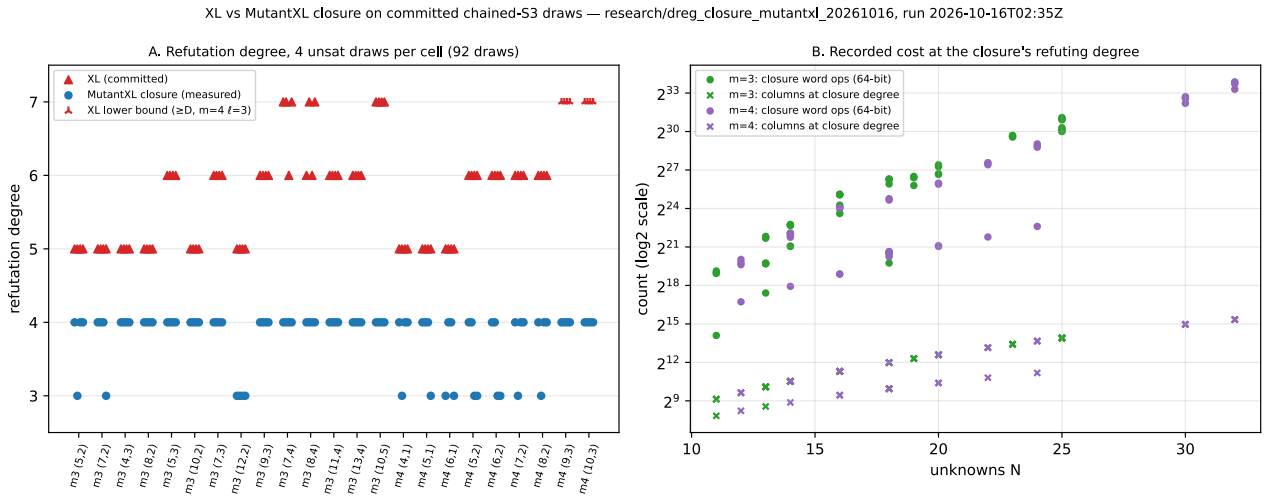


Figure 1: 92 committed unsatisfiable draws, four per cell, run 2026-10-16T02:35Z. A: XL degree (red) and closure degree (blue). B: the closure's recorded columns and counted word ops at its refuting degree.

Prediction	registered	outcome	confidence	evidence
Q1: the closure lowers the degree	every draw below XL in $\geq \frac{2}{3}$ of testable cells	**holds**: 20 of 20, plus 3 exploratory	about 55%	`score-output.txt`
Q2: by one degree	gaps of 1	**refuted**: 26 of 1, 58 of $\geq$ 2	low	same
Q3: grows with ℓ (m = 3)	medians non-decreasing, one rise	**refuted**: 4, 4, 4, 4	moderate–high	same

- **Instrument checks.** XL identity held on all 92 draws, and every closure saturated. The constructed positive control gives closure 3, XL 4.
- **Soundness control** (added after measurement): 0 of 345 satisfiable draws refuted by XL or the closure, all built to degree 5 (one above every closure refutation). Run in two invocations: the first was killed by the session tool's background time limit after 315 complete draws; the remaining 30 ran to exit 0 (`satisfiable-control-2`, peak 12.3 GB). The kill is a tool limit, not evidence.

What it means at ECC2K-130 (derived)

m	degree model	w	ℓ	$\log_2$ total	vs rho
4	XL fit $\ell + 4$	1	4	173.13	+112.32
4	**4, closure (constant)**	1	33	**68.43**	**+7.62**
4	4, closure (constant)	2	34	93.91	+33.10
3	4, closure (constant)	1	40	82.27	+21.46

- **The extrapolation.** These rows assume the closure stays at 4 from $N \leq 32$ to $N \approx 400$, which is not measured.
- **No row beats rho.** The m = 4 free-oracle floor still leaves $2^{5.71}$ per target.
- **Units are unconverted.** Column touches against group operations, with the conversion factor unmeasured.

- **Successor.** A sparse degree-4 closure at $N \approx 40$–50.